ROQ: a new measure for more accurately filtering mapped reads

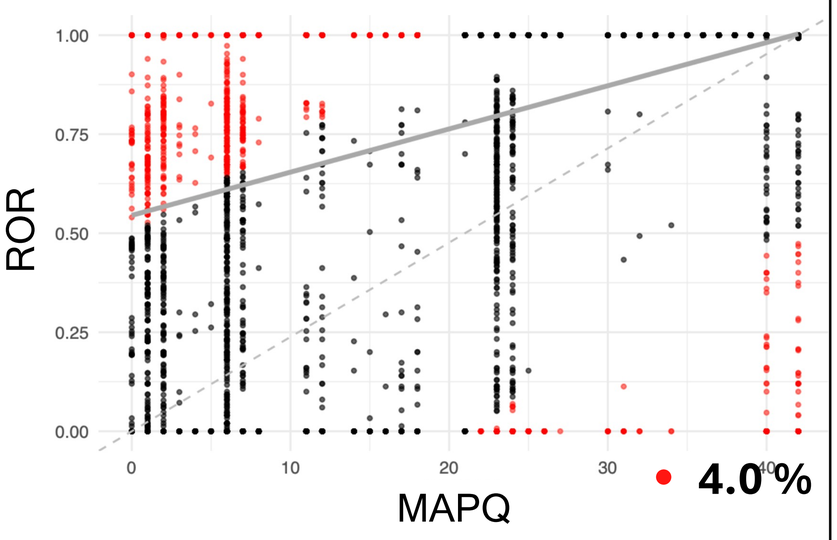
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Problem

- Next-generation sequencing (NGS) is essential for genomic research, but accurate read alignment remains challenging.
- The widely used Mapping Quality (MAPQ) score often fails to reflect true read origin—especially in complex genomic regions—leading to errors in downstream analyses.
- A more robust, origin-aware metric is needed to improve alignment quality assessment and the reliability of genomic insights.

MAPQ vs ROR



Approach

ROQ simulation data generation workflow

Read simulation

- **True read origin**

Reference genome



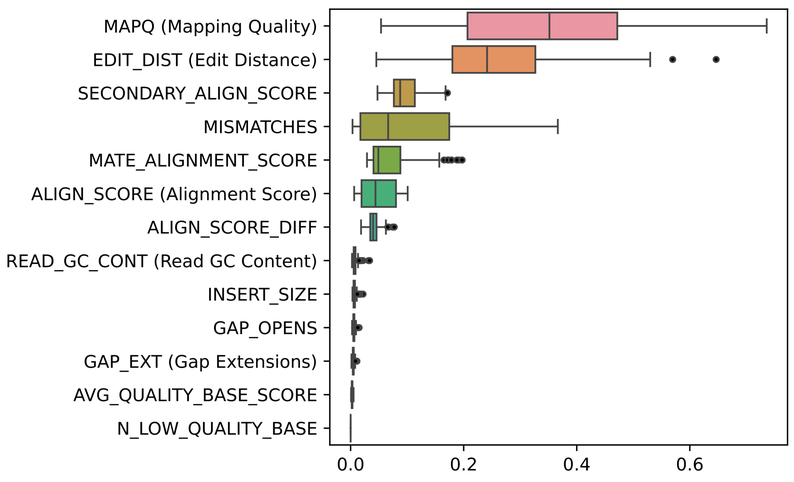
Read Alignment

- **Aligned position**
- **Alignment features**

$$Read\ Overlap\ Ratio\ (ROR) = \frac{Overlap\ length}{read\ length}$$

Alignment features → **ROQ** Machine learning model → **ROQ score (Predicted ROR)**

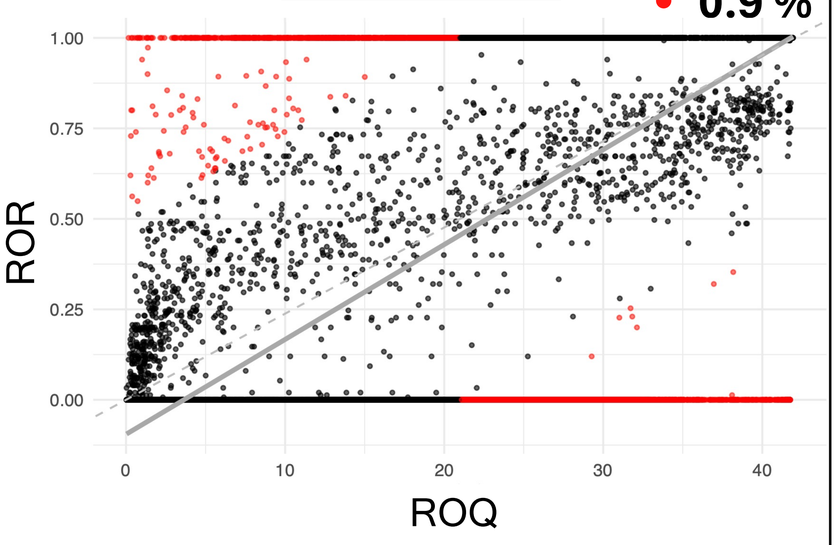
Feature Importance

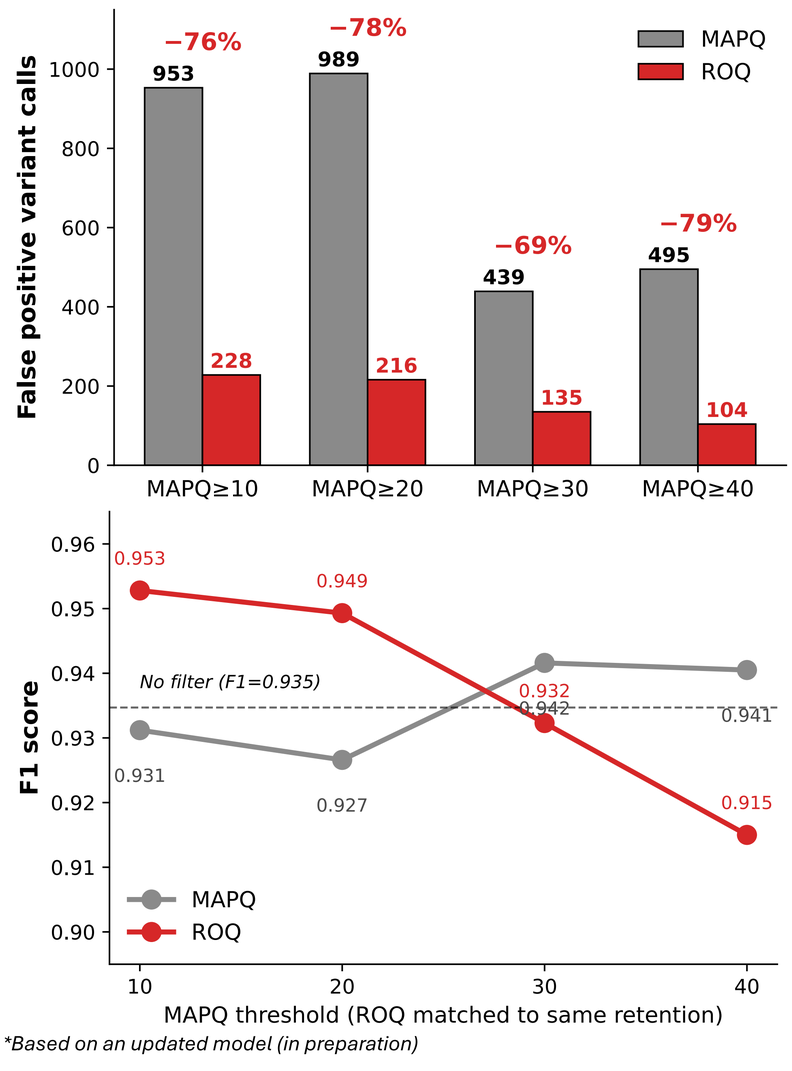


Various machine learning models

Machine learning algorithms	MAE	Corr. coef.
Linear Regression	0.0396	0.6938
Ridge Regression	0.0396	0.6938
Lasso Regression	0.0569	0.5751
Decision Tree Regressor	0.0153	0.7497
Random Forest Regressor	0.0154	0.8277
XGBoost Regressor	**0.0158**	**0.8543**

ROQ vs ROR





**Based on an updated model (in preparation)*

KEY Results

- ROQ more accurately captures true read origin than MAPQ
- Robust performance across genomic contexts, species, and different aligners
- On real GIAB NA12878 data:
  - Reduced false-positive variant calls by up to 79% at matched read retention
  - Maintained comparable or improved F1

➔ ROQ provides a robust, origin-aware complement to MAPQ for more reliable read filtering and downstream genomic analysis.

Reference Cline E, Wisittipanit N, Boongoen T, Chukeatirote E, Struss D, Eungwanichayapant A. Recalibration of mapping quality scores in Illumina short-read alignments improves SNP detection results in low-coverage sequencing data. PeerJ. 2020 Dec 7;8:e10501. doi: 10.7717/peerj.10501. PMID: 33354434; PMCID: PMC7727374.

Acknowledgements This work was supported by the Ministry of Science and ICT (NRF-2022R1F1A1065159, RS-2024-00407469 and RS-2026-25469184).